CSFD-AFM: Multimodal Remote Sensing Image Registration via Cyclic Shift Feature Descriptor and Adaptive Feature Matching Strategy

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***Abstract*—Robust registration of multimodal remote sensing images (MRSIs) remains a challenge due to nonlinear intensity differences (NIDs) and geometric distortions. Existing methods typically rely on maximum index maps (MIMs) to construct feature descriptors. However, MIM is sensitive to rotation variations, under which circumstances the number of correct matches (NCMs) decreases under severe geometric distortions. To address these problems, this study proposes an ingenious registration method (named as CSFD-AFM) based on a cyclic shift feature descriptor (CSFD) and an adaptive feature matching (AFM). CSFD is a rotation-invariant descriptor that enables efficient orientation adjustment through cyclic shifts of a 3-D feature array constructed from a modified MIM. To achieve rotation invariance, a dominant-orientation-based method for cyclic shift is introduced. Based on CSFD, we devise a matching strategy named AFM, which adaptively adjusts the scale and orientation of CSFDs for robust and accurate feature matching. Experiments were conducted on six MRSI datasets to evaluate the performance of the proposed CSFD-AFM. The results suggest that the proposed CSFD-AFM outperforms the existing seven state-of-the-art (SOTA) methods, including ReDFeat, MINIMA-LG, RIFT2, MS-HLMO, HOWP, POS-GIFT, and GLS-MIFT. In addition, our proposed CSFD-AFM achieves the accuracy within 2 pixels while maintaining its robustness against severe geometric distortions such as large rotations.**

***Index Terms*—Adaptive feature matching (AFM) strategy, cyclic shift feature descriptor (CSFD), geometric distortions, image registration, multimodal remote sensing images (MRSIs).**

I. Introduction

With the advancement of remote sensing technology in recent years, multimodal remote sensing imaging techniques, such as hyperspectral, infrared, and radar have found extensive applications in diverse domains, including but not limited to change detection [1], image fusion [2], image translation [3], target detection [4], and classification [5]. In contrast to single-modal observations, multimodal remote sensing images (MRSIs) integrate information with complementary characteristics, enabling the acquisition of more reliable observational outcomes [6]. Being defined as the process of establishing spatial correspondences between images of the same scene obtained under varying conditions (i.e., different times, viewpoints, or sensor modalities) [7], image registration serves as a crucial task in MRSI analysis. As a prerequisite for image fusion, robust and high-precision image registration is essential to ensure the effectiveness of multimodal image joint processing.

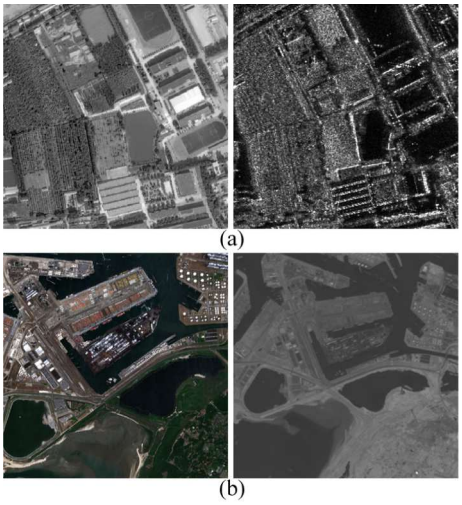


Fig. 1. Two pairs of typical MRSIs. (a) Pairwise optical–SAR images. (b) Pairwise optical–NIR images.

Despite that numerous methods have been proposed during the past decade [8], achieving reliable and accurate image registration for MRSIs remains a persistent challenge due to diverse factors. First, nonlinear intensity differences (NIDs) caused by different imaging mechanisms degrade the consistency of various modalities. Second, the characteristics of different sensors may cause geometric distortions (e.g., translation, rotation, and scale), which result in substantial reduction in the registration performance. For instance, consider the pair of optical–synthetic aperture radar (SAR) images shown in Fig. 1(a). These images exhibit noticeable NIDs, particularly the speckle noise in the SAR image, which degrades the consistency between the optical and SAR images. Fig. 1(b) illustrates a pair of optical–NIR images affected by rotation and scale variations. These distortions further lead to a decline in the number of correct matches (NCMs), compromising the robustness of registration. Therefore, it is of critical necessity to develop

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image registration methods with enhanced robustness to address these problems.

The existing MRSI registration methods can be generally categorized into three classes: area-based, feature-based, and learning-based approaches [9]. Area-based methods identify correspondences through a template matching strategy, which finds the area with the highest similarity as the correct correspondence. This kind of method is characterized by similarity measurement, and the commonly used similarity measures in the spatial domain are the sum of squared differences (SSD) [10], normalized cross correlation (NCC) [11], and mutual information (MI) [12]. Within frequency domain, the phase congruency is the most frequently adopted similarity metric due to its intensity invariance [13], [14], [15]. Ye et al. [16] proposed a novel pixelwise feature representation named channel features of orientated gradients (CFOG) to address NIDs between multimodal images. A similarity measure based on fast Fourier transform (FFT) is also developed to improve the efficiency of template matching. Zhu et al. [17] proposed a novel structural feature descriptor based on Gabor filter banks to realize registration of aerial images and light detection and ranging (LiDAR) data. Although area-based methods achieved considerably high accuracy, their sensitivity to scale and rotation differences demands information a priori to coarsely eliminate the geometric distortions.

In contrast, feature-based methods are more robust and unsusceptible to geometric distortions, which primarily include three steps: feature detection, feature description, and feature matching. One of the representative feature-based methods is the scale-invariant feature transform (SIFT) [18], which has been extensively utilized in the scope of optical image matching applications due to its robustness against scale, rotation, and illumination variations. In order to adapt to NIDs across multimodal image pairs, improved methods based on SIFT have been proposed. Dellinger et al. [19] proposed an SIFT-like descriptor adapted to SAR images based on a new gradient computational robust to speckle noise. Later, Xiang et al. [20] developed the OS-SIFT method, which utilized two different operators to establish gradient consistency between SAR and optical images. Although these improved methods yield satisfactory results for specific image types, however, they are generally not applicable to other image types, for which reason their practical utility is severely limited in multimodal scenarios. To enhance the robustness of the descriptors across different modalities, a variety of descriptors based on local self-similarity were developed [21], [22], [23]. Local self-similarity captures the self-similarity patterns within a log-polar spatial structure to quantify the internal structural correlations amongst multimodal images. Based on this, Xiong et al. [24] proposed a feature description method utilizing orientations of self-similarities with a grid structure inspired by gradient location and orientation histogram (GLOH) [25]. This approach demonstrates superior performance across optical, SAR, and LiDAR images.

In recent studies, the phase congruency has emerged as a prominent feature-based approach due to its robustness against modality variations across diverse image types. In 2018, based on the phase congruency model, Ye et al. [26] devised an effective local feature, which is invariant to illumination variation. Li et al. [27] proposed the maximum index map (MIM) based on phase congruency for feature description. MIM is constructed from the log-Gabor convolution sequence, which is much more robust against NIDs than a conventional gradient map. However, this method relies on loop traversal of the log-Gabor convolution sequence to eliminate the effect of rotations, which markedly deteriorates its efficiency and robustness. In response to this shortcoming, Zhu et al. [28] proposed the R2FD2 by introducing a rotation-invariant RMIM, which is derived by first adjusting the values of the MIM and then rotating the adjusted MIM. Hou et al. [29] devised the POS-GIFT method to realize the rotation invariance of log-Gabor convolution sequence. This approach first reorders the elements within surrounding feature vectors based on the dominant orientation and then concatenates them in a clockwise sequence starting from that orientation. Finally, the central feature vector is appended to form the complete descriptor. Fan et al. [30] proposed the GLS-MIFT method, which employs a similar rotation-invariant mechanism to POS-GIFT. It involves reordering the elements within feature vectors and concatenating them in a clockwise sequence. However, it is distinguished by the use of a novel sampling grid based on the radial partition expansion. It must be noted that assigning dominant orientations to feature points and constructing feature descriptors according to these orientations may introduce substantial mismatches, which greatly deteriorates the performance of registration.

Recent years have witnessed substantial advancements in deep learning technology, particularly in the field of remote sensing, where learning-based methods have evolved into an indispensable component. Li et al. [31] proposed the semantic position probability distribution (SPPD), a framework of semantic position probability distribution for SAR–optical image matching. Gao et al. [32] proposed SFA-Net, a segment anything model (SAM)-guided attention network designed to address the challenge of multimodal image registration under severe nonlinear distortions. Hughes et al. [33] devised a three-step deep learning framework for SAR–optical image registration, where candidate feature proposal, feature matching, and outlier rejection are encoded by dedicated deep neural networks, respectively. With respect to feature detection, Sun et al. [34] developed a detector-free matching framework that establishes accurate matches with Transformers in a coarse-to-fine manner. In more challenging cross-modal scenarios, several recent deep learning approaches have emerged. Deng and Ma [35] proposed the ReDFeat, which recouples the detection and description through a mutual weighting strategy to stabilize end-to-end cross-modal feature learning. Ren et al. [36] reported MINIMA, a unified framework that achieves robust cross-modal image matching by leveraging generative models to scale up training data from RGB sources.

Although deep learning techniques exhibit strong potential in the field of image registration, learning-based methods are deteriorated by the training data problems. Specifically, the trained model exhibits dependency on the training data, leading to severe performance degradation when applied to unseen

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modalities. In addition, there is a lack of comprehensive and diverse datasets to generate generalizable networks, which hinders the wide application of learning-based methods.

Despite that extensive efforts have been made to improve the robustness and accuracy of the registration methods, NCMs significantly decline under rotation and scale variations. To deal with these problems, this study proposes a feature-based method named cyclic shift feature descriptor adopting adaptive feature matching (CSFD-AFM) strategy to improve the registration performance under severe NIDs and geometric distortions. First, the feature points are detected in scale spaces through a maximum moment map analysis. Subsequently, a modified MIM termed high-density MIM HMIM is constructed as the intermediate modal for description. Based on HMIM, a 3-D feature array with a cyclic shift function is proposed to permute orientation indices and rotate the coordinate system, thereby realizing orientation adjustment. The cyclic shift vector is determined by a dominant orientation and then used to adjust the feature array. CSFD is constructed by dimensionality reduction and vectorization on the adjusted feature array. Furthermore, an AFM strategy is performed, which utilizes the transformation derived by initial matching results to adjust local neighborhood scale of feature points and regenerate the feature arrays. The transformation also narrows the matching range, within which the cyclic shift vector is determined via cross correlation between feature arrays to achieve orientation alignment. The feature array after adjustment undergoes vectorization, forming CSFD. Finally, we employ nearest neighbor distance matching to obtain matches, which is then refined using the fast sample consensus (FSC) algorithm [37] to refine the matches and obtain the transformation between images. The key contributions of this study are summarized as follows.

1) A new feature description format in the form of a 3-D feature array is proposed to unify orientation index permutation and coordinate system rotation into a cyclic shift function, enabling efficient orientation adjustment.
2) A CSFD is proposed, which employs cyclic shift vectors on feature arrays to achieve rotation invariance. The cyclic shift vectors can be constructed through two approaches: dominant orientation approach and cross-correlation approach.
3) A novel AFM strategy is proposed, which leverages the geometric transformation estimated from initial matching results to adjust the configuration of CSFDs.

The remainder of this article is organized as follows. Section II outlines the whole process of the proposed method, including the feature detection, the 3-D feature array, CSFD, and AFM strategy. Section III presents the experiments and analysis. Section IV summarizes the conclusion of this article.

II. Methodology

In this section, the proposed CSFD-AFM is elaborated in detail. First, we extract feature points from scale spaces. Second, we propose a novel feature descriptor CSFD, which achieves rotation invariance through cyclic shift of a 3-D feature array. Finally, an AFM strategy is performed for feature matching. The flowchart of our proposed method is demonstrated in Fig. 2.

A. Feature Detection

It is necessary to detect sufficient feature points with uniform spatial distribution amongst MRSIs for transformation evaluation. Due to the absence of the scale factor amongst input images, the scale spaces of the input images are constructed to deal with the scale variations. The scale space is constructed as a stack of images derived from the original image scaled by different factors. The scaling factors comprise two sequences k_1 and k_2, which can be presented as follows:

$$\left\{k_1 = \frac{2}{3} \cdot \left(\frac{1}{2}\right)^n | n \in N, k_1 \cdot L > 2 \cdot R\right\} \tag{1}$$

$$\left\{k_2 = \left(\frac{1}{2}\right)^n | n \in Z^+, k_2 \cdot L > 2 \cdot R\right\} \tag{2}$$

where L is the smaller of the image width and height and R is the radius of the local neighborhood centered at a feature point. The constraints $k_1 \cdot L > 2 \cdot R$ and $k_2 \cdot L > 2 \cdot R$ ensure that the downsampled images are larger than the feature point neighborhood, resulting in two finite sets of scaling factors. The elements from these two sets are combined into a single set and sorted in descending order to form the final sequence of scaling factors. The original image is scaled by each factor in this sequence to generate images at different resolutions, which collectively constitute the scale space. In this study, we extract the maximum moment maps [27] of each layer in the scale space first and then detect feature points on them through the features from accelerated segment test (FAST) [38] algorithm. Previous studies [39], [40] have demonstrated that the maximum moment map based on phase congruency effectively preserves the salient structures across diverse image modalities, enabling a certain degree of consistent feature detection. Fig. 3 presents the input images, the corresponding maximum moment maps, and detected feature points. It can be observed that the maximum moment maps derived from both optical and SAR images exhibit a high degree of consistency, where significantly higher values are predominantly distributed along edge regions. Accordingly, the feature points detected by FAST are spatially aligned with these prominent edge structures. The results indicate that the integration of the maximum moment map with the FAST algorithm allows for a similar spatial distribution of feature points and an adequate number of them, thereby supporting subsequent feature matching.

B. 3-D Feature Array

After performing the feature detection, feature description is conducted. MIM, which consists of orientation indices derived from the local structures, demonstrates superior consistency across diverse image modalities. Unfortunately, it is still challenging to construct the rotation-invariant descriptors on MIM. The existing methods typically assign multiple dominant orientations to achieve rotation invariance [28], [26]. For each of the dominant orientations, a local coordinate system is established, followed by the construction of corresponding descriptor. Nevertheless, this approach is confronted by two

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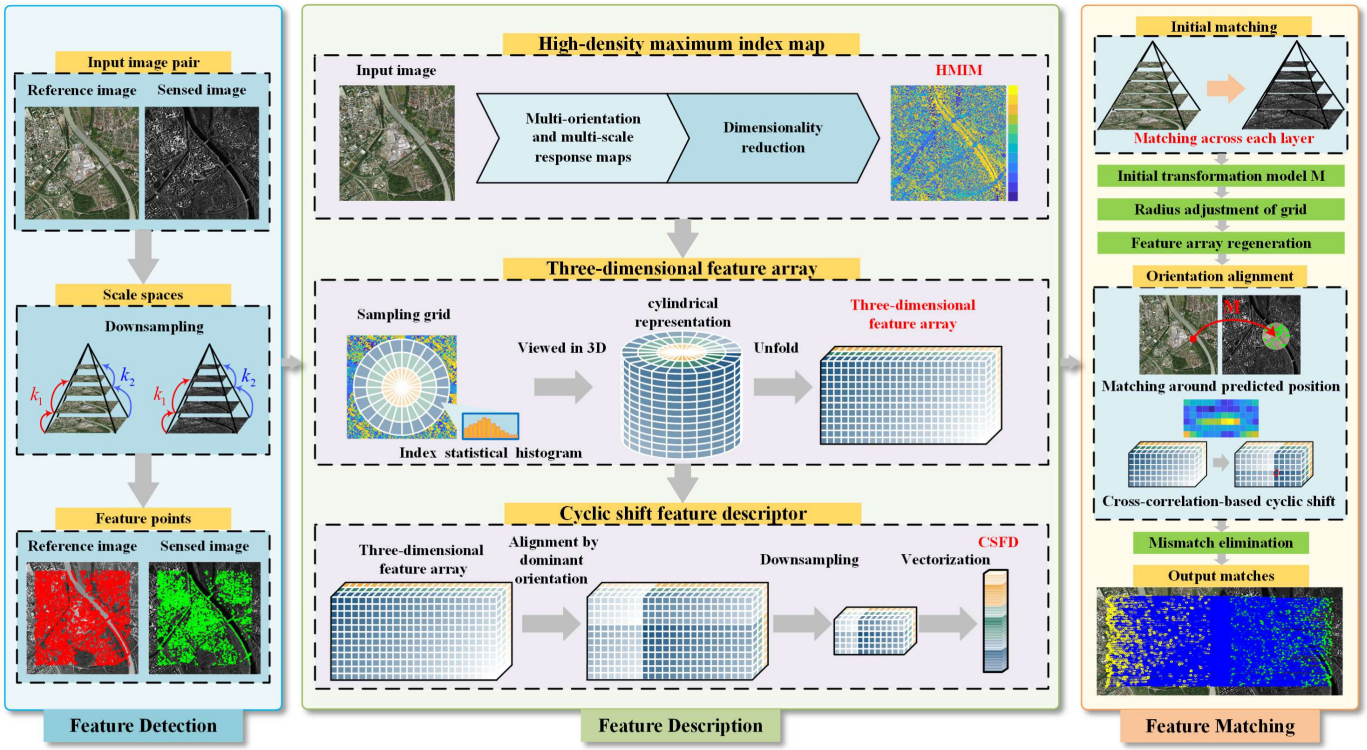


Fig. 2. Flowchart of the proposed method.

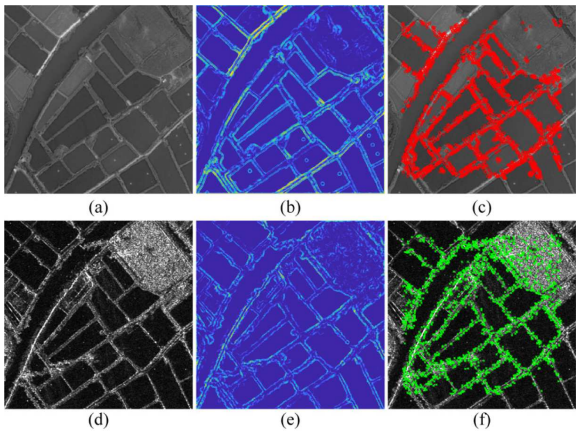


Fig. 3. Feature detection on pairwise optical–SAR images. (a) Optical image. (b) Maximum moment map of the optical image. (c) Feature points on the optical image. (d) SAR image. (e) Maximum moment map of the SAR image. (f) Feature points on the SAR image.

critical limitations. First, when multiple dominant orientations are assigned, the descriptors must be reconstructed for every orientation respectively. Second, NIDs often hamper consistent dominant orientation assignment amongst MRSIs, resulting in considerable mismatches. To solve these problems, we propose a rotation-invariant feature descriptor named CSFD, based on a novel description format: the 3-D feature array. We first extend MIM by increasing the number of orientation indices to form HMIM and then construct the 3-D feature array with orientation-adjustment capability. The cyclic shift function is also proposed for orientation adjustment. The details of the 3-D feature array are provided as follows.

To enhance rotation invariance, we implement a modified MIM termed HMIM through orientation density augmentation. Specifically, the HMIM is constructed using a bank of steerable filters [41] to generate convolution response maps across multiple orientations and scales, followed by dimensionality reduction. This process is presented in Fig. 4. Unlike the log-Gabor filter used in MIM, we employ a simpler yet effective steerable filter bank to derive HMIM. The steerable filter bank is a set of oriented filters with the property that a filter at an arbitrary orientation can be synthesized from linear combinations of the basis filters. This approach offers two primary advantages. First, the steerable filters operate in the spatial domain directly. They generate multiorientation and multiscale convolutional responses comparable to those of the log-Gabor filter while avoiding the even-symmetric and odd-symmetric wavelets required in the log-Gabor approach. Second, the filters for any desired orientation can be efficiently generated as linear combinations of basis filters. This property facilitates orientation density augmentation of HMIM. The first-order partial derivatives of the Gaussian function are pairwise simple and widely used basis filters, which can be expressed as follows:

$$G_\sigma(x,y) = \frac{1}{2\pi\sigma^2}\exp\left(-\frac{x^2+y^2}{2\sigma^2}\right) \tag{3}$$

$$\frac{\partial G_\sigma}{\partial x} = -\frac{1}{2\pi\sigma^4}x\exp\left(-\frac{x^2+y^2}{2\sigma^2}\right) \tag{4}$$

$$\frac{\partial G_\sigma}{\partial y} = -\frac{1}{2\pi\sigma^4}y\exp\left(-\frac{x^2+y^2}{2\sigma^2}\right) \tag{5}$$

where $G_\sigma(x,y)$ is the Gaussian function and σ is the standard deviation of $G_\sigma(x,y)$. The terms $(\partial G_\sigma)/(\partial x)$ and $(\partial G_\sigma)/(\partial y)$

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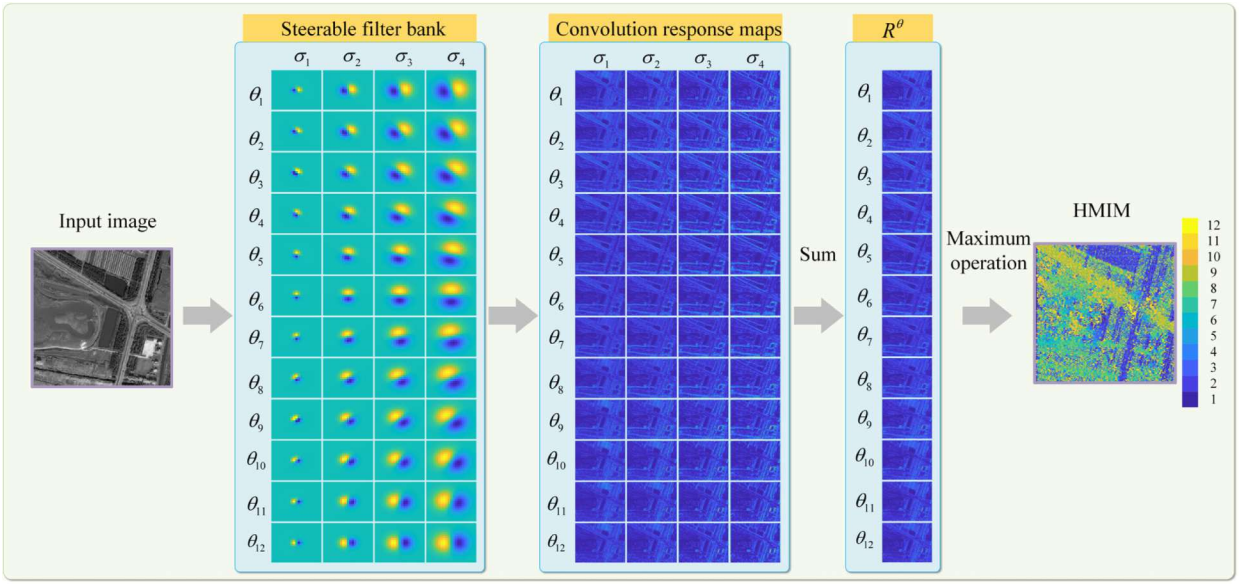


Fig. 4. Construction process of HMIM.

denote the first-order partial derivatives. Based on (4) and (5), the filter $G_{\sigma_i}^{\theta_j}(x,y)$ at scale σ_i and orientation θ_j can be synthesized by

$$G_{\sigma_i}^{\theta_j}(x,y) = \left[\cos\left(\theta_j\right), \sin\left(\theta_j\right)\right] \begin{bmatrix} \frac{\partial G_{\sigma_i}}{\partial x} \\ \frac{\partial G_{\sigma_i}}{\partial y} \end{bmatrix} = -\frac{1}{2\pi\sigma_i^4}\left[\cos\left(\theta_j\right)x + \sin\left(\theta_j\right)y\right]\exp\left(-\frac{x^2+y^2}{2\sigma_i^2}\right) \tag{6}$$

where $\sigma_i = \tau \cdot i, i \in [1, N_\sigma]$ and $\theta_j = \pi(j-1)/N_\theta, j \in [1, N_\theta]$. N_σ and N_θ denote the numbers of orientations and scales, respectively, τ denotes a scale coefficient. Typically, the parameters N_σ and τ are set to 4 and 0.8, respectively. In this work, N_θ is set to 12, which is twice the conventional value used in existing methods, to enhance orientation sensitivity. Subsequently, the convolutional response map $R_{\sigma_i}^{\theta_j}(x,y)$ is obtained by

$$R_{\sigma_i}^{\theta_j}(x,y) = \text{abs}\left[G_{\sigma_i}^{\theta_j}(x,y) * I(x,y)\right] \tag{7}$$

where the operator * indicates the convolutional calculation, the function abs[·] is used to calculate the absolute value, and $I(x,y)$ is the grayscale version of the input image.

The convolutional response maps incorporate redundant components for feature description, for which reason dimensionality reduction is applied to obtain HMIM. For each orientation θ_j, the response maps of all N_σ scales are first summed to obtain the orientation-specific response map $R^{\theta_j}(x,y)$. Then, a total of N_θ orientation-specific response maps are stacked to form the multiorientation response array $R^{\theta}(x,y)$. For each pixel location (x_k, y_k) of the input image, we can obtain a vector $\{R^{\theta_j}(x_k,y_k)|j \in [1,N_\theta]\}$ in $R^{\theta}(x,y)$. After finding the maximum value and its corresponding index $j_{\max}$ in this vector, we set $j_{\max}$ as the value of the location (x_k, y_k)

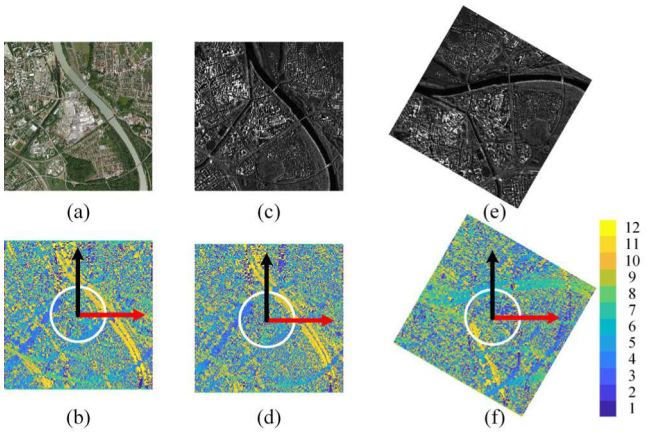


Fig. 5. Effect of rotation on HMIM. (a) Optical image. (b) HMIM of (a). (c) SAR image. (d) HMIM of (c). (e) SAR image with rotation. (f) HMIM of (e). White circles represent the local neighborhoods. Red and black arrows represent the local coordinate systems.

in HMIM. The dimensionality reduction process elaborated above can be expressed as follows:

$$R^{\theta_j}(x,y) = \sum_{i=1}^{N_\sigma} R_{\sigma_i}^{\theta_j}(x,y) \tag{8}$$

$$\text{HMIM}(x_k,y_k) = \arg\max_j \left[R^{\theta_j}(x_k,y_k)\right], \quad j \in [1,N_\theta] \tag{9}$$

where the operator arg max[·] is used to obtain the index of the maximum in $\{R^{\theta_j}(x_k,y_k)|j \in [1,N_\theta]\}$.

As shown in Fig. 5(a)–(d), HMIMs of optical and SAR images exhibit similar index distributions, which verifies the consistency of HMIM across different modalities. However, when the image is rotated, this consistency degrades. As shown in Fig. 5(c)–(f), the corresponding terrain features in the two images with rotation variations are assigned with different orientation indices. In addition, the coordinate axes of the local neighborhoods in two images are oriented in

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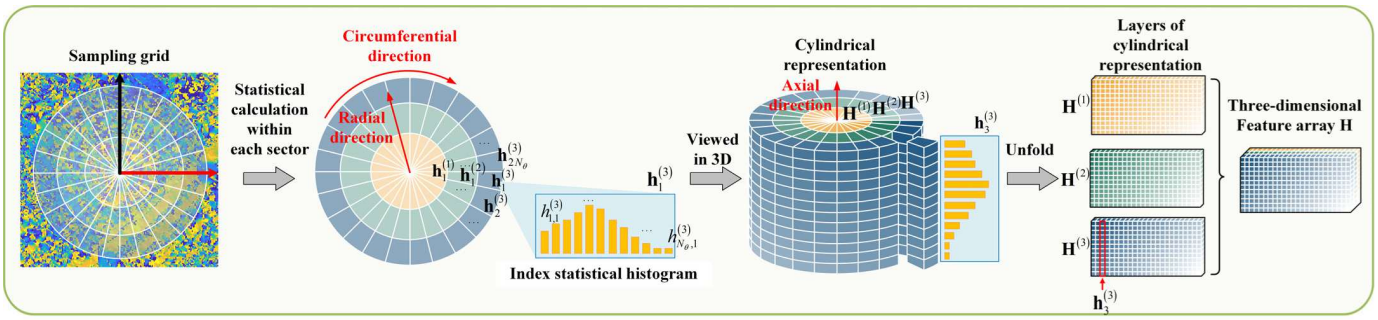


Fig. 6. Construction process of the 3-D feature array.

different terrain-referenced directions, suggesting that the corresponding terrain information is encoded at different locations within the descriptors. To address these problems, we therefore construct a 3-D feature array to achieve orientation-adaptive representation. The construction process is presented in Fig. 6. First, a regular centrosymmetric sampling grid is adopted within the local neighborhood centered at each feature point on HMIM. Each local neighborhood is defined as a circular region with a constant radius R. The grid divides the local neighborhood into N_r concentric circular layers with uniform radial spacing. The default value of N_r is 3. Each layer is partitioned into $2 \times N_\theta$ equiangular sectors along the circumferential direction. Therefore, the local neighborhood surrounding the feature point is partitioned into a total of $N_r \times 2 \times N_\theta$ sectors. Then, the index statistical histograms with N_θ bins are calculated for each sector. To elucidate this structure, the histograms from each sector are vectorized and axially stacked into a cylindrical representation, which is subsequently unfolded, arranging each concentric layer into a $N_\theta \times 2N_\theta$ matrix

$$\mathbf{H}^{(i)} = \left[\mathbf{h}_1^{(i)}, \cdots \mathbf{h}_m^{(i)}, \ldots, \mathbf{h}_{2N_\theta}^{(i)}\right] \begin{bmatrix} h_{1,1}^{(i)} & h_{1,2}^{(i)} & \cdots & \cdots & h_{1,2N_\theta}^{(i)} \\ h_{2,1}^{(i)} & \ddots & \vdots & \cdot & \vdots \\ \vdots & \cdots & h_{n,m}^{(i)} & \cdots & \vdots \\ \vdots & \cdot & \vdots & \ddots & \vdots \\ h_{N_\theta,1}^{(i)} & \cdots & \cdots & \cdots & h_{N_\theta,2N_\theta}^{(i)} \end{bmatrix}, \quad i \in [1, N_r] \tag{10}$$

where $\mathrm{h}_m^{(i)}$ denotes the mth index statistical histogram in the ith matrix $\mathrm{H}^{(i)}$ and $h_{n,m}^{(i)}$ denotes the nth element in $\mathrm{h}_m^{(i)}$. Finally, the matrices are stacked to form the 3-D feature array as follows:

$$\mathrm{H} = \left\{\mathrm{H}^{(1)}, \mathrm{H}^{(2)}, \ldots, \mathrm{H}^{(N_r)}\right\} \tag{11}$$

where $\mathbf{H}$ is the 3-D feature array.

To clarify the process described above, the pseudo-code for extracting the 3-D feature array is presented in Algorithm 1.

One of the key advantages of the 3-D feature array lies in its capability to realize orientation alignment through cyclic shifts. The cyclic shift method is grounded in the observation that although the rotation alters the index values at corresponding positions on HMIMs, the statistical distribution of these index values remains invariant. For example, the largest river in Fig. 5(c), which corresponds to indices 11 and 12 in Fig. 5(d), undergoes a rotation transformation that leads to the corresponding indices being collectively shifted to indices 7 and 8 in Fig. 5(f). Specifically, the rotation actually produces a cyclic permutation of orientation indices in the HMIM, where each index i is updated as follows:

$$i' = (i + \Delta N) \bmod N_\theta \tag{12}$$

where ΔN denotes the rotation-dependent index increment, N_θ denotes the maximum index value, and the symbol "mod"

Algorithm 1 Extraction of 3-D Feature Array

Require: $image, filter_bank, N_\theta, N_\sigma, feature_points, R, N_r$
Ensure: $3D_feature_arrays$

```
I ← convert_to_grayscale_if_color (image) // dimensions: [H, W]
R ← zeros(N_θ, N_σ, H, W)
for (o, s) ∈ [0, N_θ − 1] × [0, N_σ − 1] do
    R[o, s, :, :] ←| convolution (I, filter_bank [o, s]) | // (7)
end for
R_sum ← zeros(N_θ, H, W)
for o = 0 to N_θ − 1 do
    ori_resp ← R[o, :, :, :]   // dimensions: [N_σ, H, W]
    R_sum [o, :, :] ← sum(ori_resp, axis = 0) // (8)
end for
HMIM ← zeros(H, W)
HMIM ← arg max_k R_sum[k, :, :] // (9)
3D_feature_arrays ← empty list
for each point in feature_points do
    patch ← extract_neighborhood (HMIM, point, R)
    feature_3d ← zeros(N_r, N_θ, 2N_θ) // a 3D feature array
    for (r, a) ∈ [0, N_r − 1] × [0, 2N_θ − 1] do
        // (r, a): radial and angular indices of polar coordinate partition
        region ← get_polar_sector (patch, r, a, N_r, 2N_θ)
        histogram ← compute_histogram (region, N_θ)
        // N_θ defines number of histogram bins
        feature_3d[r, :, a] ← histogram // (10) (11)
    end for
    append 3D_feature_arrays with feature_3d
    // 3D feature arrays of feature points
end for
return 3D_feature_arrays
```

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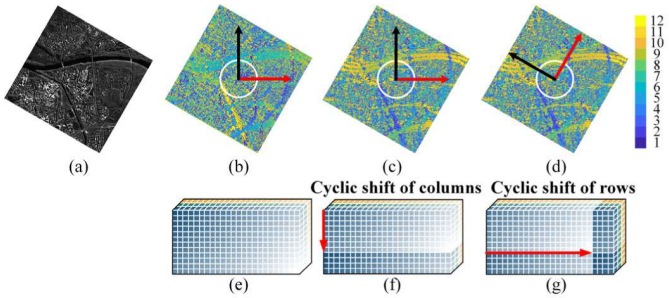


Fig. 7. Mechanism of cyclic shift on the 3-D feature array. (a) SAR image with rotation. (b) HMIM of (a). (c) HMIM after cyclic permutation of orientation indices. (d) HMIM after rotation of coordinate system. (e) Feature array in (b). (f) Feature array in (c). (g) Feature array in (d).

denotes the modulo operation that returns a positive result. This cyclic permutation of orientation indices in the HMIM functions as a cyclic shift of index statistical histogram $\mathrm{h}_m^{(i)}$, which can be achieved by a vertical cyclic shift of the column vectors in each matrix of $\mathbf{H}$. In the same way, the rotation of the coordinate system functions as a rotation of concentric circular layers in the cylindrical representation, which can be achieved by a horizontal cyclic shift of the row vectors in each matrix of $\mathbf{H}$. This process is illustrated in Fig. 7. Within each matrix of $\mathbf{H}$, a cyclic permutation of the orientation indices corresponds to a cyclic shift of the column vectors, and a rotation of the coordinate system corresponds to a cyclic shift of the row vectors, respectively. As shown in Fig. 7(d), within the local neighborhood, HMIM after being adjusted exhibits consistency with HMIM presented in Fig. 5(b), to which their coordinate axes consistently oriented toward the same terrain features. For ease of expression, we define the 2-D cyclic shift function $\psi(\mathrm{H}^{(i)}, (u, v))$ to achieve orientation adjustment as follows:

$$\psi\left(\mathbf{H}^{(i)}, (u, v)\right) = \begin{bmatrix} h'^{(i)}_{1,1} & h'^{(i)}_{1,2} & \cdots & \cdots & h'^{(i)}_{1,2N_\theta} \\ h'^{(i)}_{2,1} & \ddots & \vdots & \iddots & \vdots \\ \vdots & \cdots & h'^{(i)}_{n,m} & \cdots & \vdots \\ \vdots & \iddots & \vdots & \ddots & \vdots \\ h'^{(i)}_{N_\theta,1} & \cdots & \cdots & \cdots & h'^{(i)}_{N_\theta,2N_\theta} \end{bmatrix} \tag{13}$$

$$h'^{(i)}_{n,m} = h^{(i)}_{(n-u) \bmod N_\theta, (m-v) \bmod 2N_\theta} \tag{14}$$

where (u, v) denotes the cyclic shift vector, with u and v being the row and column offsets, respectively. The parameters u and v are restricted to the sets $\{1, 2, \ldots, N_\theta - 1\}$ and $\{1, 2, \ldots, 2N_\theta - 1\}$, respectively. The term $h'^{(i)}_{n,m}$ denotes the element after a cyclic shift in the nth row and mth column.

This function implements a bidimensional cyclic shift operation on each matrix of $\mathbf{H}$, where the element $h^{(i)}_{1,1}$ is moved to replace the element $h^{(i)}_{(1+u,1+v)}$. The feature array and cyclic shift operation offer two distinctive advantages. First, the feature arrays corresponding to different dominant orientations can be efficiently derived from cyclic shifts on the original feature array. Second, the optimal cyclic shift vector minimizing the Euclidean distance between two feature arrays can be determined by cross correlation.

C. Cyclic Shift Feature Descriptor

In this section, we introduce a dominant-orientation-based cyclic shift method for adjusting feature arrays, which generates multiple CSFDs per feature point for different dominant orientations. Fig. 8 demonstrates the process of this method.

First, we construct an overall index statistical histogram by summing all index statistical histograms $\mathrm{h}_m^{(i)}$ in the feature array $\mathbf{H}$. The overall index statistical histogram counts the index distribution within the local neighborhood. The index corresponding to the peak in the overall index statistical histogram is set as the dominant index (DI).

Second, we calculate the principal direction to modify DI for the cyclic shift of row vectors in each matrix of the feature array. As expressed in (4) and (5), the orientations of the steerable filters are uniformly distributed between 0 and π, and the convolutional responses are computed using absolute values. This necessitates determining whether the dominant orientation corresponds to the $[0, \pi]$ interval or its complementary direction. To resolve this ambiguity, we introduce the concept of principal direction. Specifically, the pixels associated with the dominant index within the local neighborhood are identified as the dominant pixels. For each feature point p_i, the dominant pixels collectively constitute the set $N(p_i)$. The centroid (x_c, y_c) of $N(p_i)$ is then calculated by

$$\begin{cases} x_c = \dfrac{1}{|N(p_i)|} \displaystyle\sum_{j \in N(p_i)} x_j \\ y_c = \dfrac{1}{|N(p_i)|} \displaystyle\sum_{j \in N(p_i)} y_j \end{cases} \tag{15}$$

where (x_j, y_j) denotes the Cartesian coordinates within the local neighborhood of the jth pixel and $|N(p_i)|$ denotes the cardinality of the set $N(p_i)$. The principal direction can be calculated by

$$\theta_{\mathrm{PD}} = \arctan 2\,(y_c / x_c) \tag{16}$$

where $\arctan 2(\cdot)$ is the four-quadrant arctangent function. The initial dominant orientation DI is calibrated by θ_{PD}, which can be expressed as follows:

$$\mathrm{DI}' = \begin{cases} \mathrm{DI}, & \theta_{\mathrm{PD}} \leqslant 0 \\ \mathrm{DI} + N_\theta, & \text{else} \end{cases} \tag{17}$$

where DI' denotes the calibrated dominant index.

Third, based on the derived DI and DI', each matrix in the feature array undergoes a cyclic shift of $N_\theta - \mathrm{DI}$ along its columns and a cyclic shift of $2N_\theta - \mathrm{DI}'$ along its rows. The cyclic shift of $N_\theta - \mathrm{DI}$ on column vectors shifts the dominant index to the final position N_θ in each index statistical histogram $\mathrm{h}_m^{(i)}$, thereby aligning the indices. Since the rotation of the coordinate system corresponds to row vectors in each matrix of the feature array, the cyclic shift of $2N_\theta - \mathrm{DI}'$ actually transfers the corresponding terrain information to the same position in the coordinate system. Therefore, the cyclic shift vector $(u_{\mathrm{DI}}, v_{\mathrm{DI}})$ can be expressed as follows:

$$(u_{\mathrm{DI}}, v_{\mathrm{DI}}) = \left(N_\theta - \mathrm{DI}, 2N_\theta - \mathrm{DI}'\right). \tag{18}$$

Each matrix $\mathrm{H}^{(i)}$ in the feature array is adjusted by

$$\psi\left(\mathrm{H}^{(i)}, (u_{\mathrm{DI}}, v_{\mathrm{DI}})\right), \quad i = 1, 2, 3. \tag{19}$$

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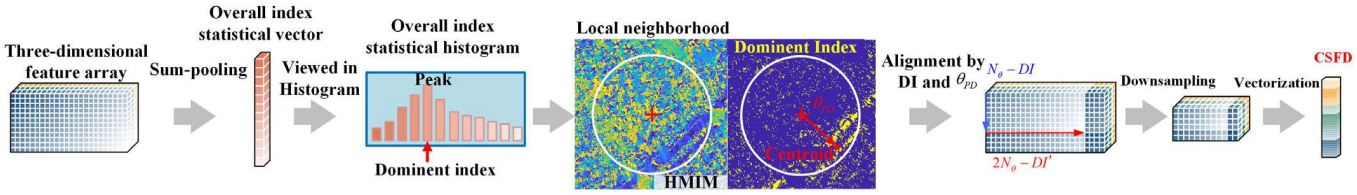


Fig. 8. Process of dominant-orientation-based cyclic shift.

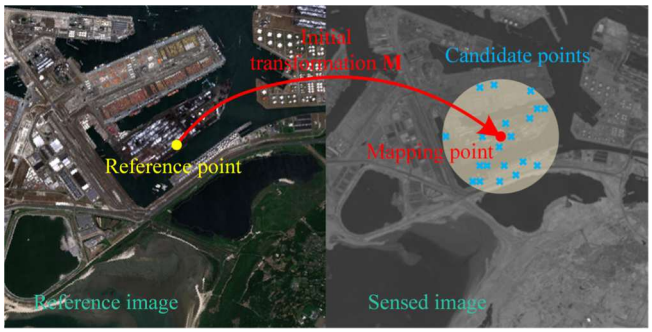


Fig. 9. Spatial constraints on the matching range.

After performing the cyclic shift adjustment, each matrix in the feature array is subsampled by removing even-indexed rows and columns to reduce descriptor dimensionality. The column vectors of each matrix in the subsampled feature array are then concatenated sequentially to form the final CSFD. Different from the existing dominant orientation assignment methods, CSFDs corresponding to other orientations can be efficiently derived by performing cyclic shift operations on the original feature array. To enhance registration robustness against rotation variations, cyclic shift vectors are additionally established using indices whose magnitudes exceed 80% of the maximum magnitude in the overall index statistical histogram. The original feature array is also adjusted by these additional cyclic shift vectors, which generate corresponding CSFDs, enabling multiple descriptors for each feature point.

D. Adaptive Feature Matching

The initial matching is performed on the derived CSFDs via global cross-layer matching within the scale spaces. We first calculate the distance between the CSFDs, and the point pair with the nearest distance of CSFDs is considered a candidate match. Then, we use the FSC algorithm to eliminate the mismatches and obtain the transformation matrix $\mathbf{M}$. The matrix $\mathbf{M}$ can be mathematically decomposed into translation, rotation, scale, and shearing components as follows:

$$\begin{aligned}\mathbf{M} &= \begin{bmatrix} A_{11} & A_{12} & b_{13} \\ A_{21} & A_{22} & b_{23} \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & b_{13} \\ 0 & 1 & b_{23} \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &\quad * \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 1 & m & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned} \tag{20}$$

where terms $A_{11}, A_{12}, A_{21}, A_{22}, b_{13}$, and b_{23} denote the elements in the matrix M, θ denotes the rotation angle, s_x and s_y denote the scale coefficient in x-direction and y-direction, respectively, and m is the shearing coefficient. By expanding the matrix equation into a system of linear equations and solving the resultant linear system, the scale coefficients expressed in terms of the elements of the matrix $\mathbf{M}$ can be derived as follows:

$$s_x = \sqrt{A_{11}^2 + A_{21}^2} \tag{21}$$

$$s_y = \frac{A_{11}^* A_{22} - A_{21}^* A_{12}}{\sqrt{A_{11}^2 + A_{21}^2}}. \tag{22}$$

Based on s_x and s_y, we calculate the scale factor by

$$s = \frac{s_x + s_y}{2}. \tag{23}$$

We reuse the feature arrays in the reference image directly for efficiency. As for the feature arrays in the sensed image, we adapt the radii of the local neighborhood regions from R to $R' = s \times R$ and recreate the feature arrays to address scale variations. By leveraging the derived transformation matrix $\mathbf{M}$, we conduct spatial constraints on the matching scopes. As shown in Fig. 9, the feature points on the reference image (denoted as reference points) are first mapped to the sensed image via a transformation matrix $\mathbf{M}$. Then, the K-nearest neighboring feature points around each mapping point on the sensed image are identified as the candidate points of the corresponding reference point. The default value of K is set to 20.

Unlike dominant-orientation-based CSFDs that determine the cyclic shift vector based on the local neighborhood of the feature point, AFM adjusts the cyclic shift vector adaptively by incorporating distinctive characteristics of the target feature to be matched. This process is presented in Fig. 10. First, each matrix in the feature array is subsampled by removing even-indexed rows and columns for efficiency. Second, the reference point and each candidate point constitute the pairwise reference and candidate points. We sum all the matrices of each feature array of the pairwise points to generate two composite matrices, which are expressed as follows:

$$\mathrm{M}_R = \sum_{i=1}^{N_r} \mathrm{H}_R^{(i)} \tag{24}$$

$$\mathrm{M}_C = \sum_{i=1}^{N_r} \mathrm{H}_C^{(i)} \tag{25}$$

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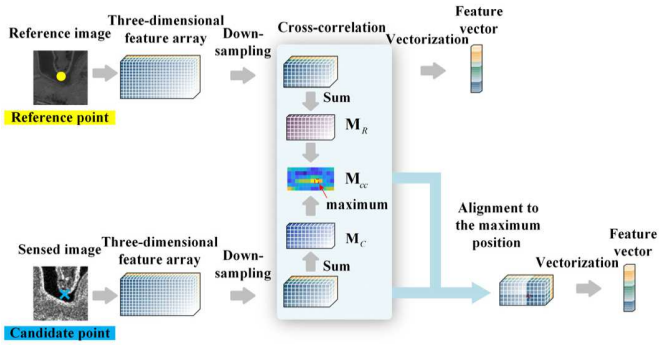


Fig. 10. Process of cross-correlation-based cyclic shift.

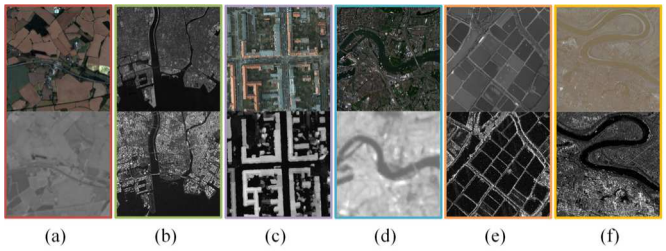


Fig. 11. Typical image pairs of the MRSI datasets. (a) Optical–NIR image pair. (b) NIR–SAR image pair. (c) Optical–DSM image pair. (d) Optical–infrared image pair. (e) Optical–SAR1 image pair. (f) Optical–SAR2 image pair.

where M_R and M_C are the composite matrices of the pairwise reference and candidate points, respectively, and $H_R^{(i)}$ and $H_C^{(i)}$ denote the matrices in the corresponding feature arrays. Based upon M_R and M_C, the cross-correlation matrix M_{cc} is calculated by using the cyclic shift function, which is expressed as follows:

$$M_{cc}(i,j) = \sum\sum\left[\psi(M_C,(i,j)) \cdot M_R\right] \tag{26}$$

where the symbol "$\sum\sum[\cdot]$" denotes the function that sums all elements in a matrix. The coordinates of the maximum value of M_{cc} are identified and designated as the cyclic shift vector (u_{cc}, v_{cc}), which can be expressed as follows:

$$(u_{cc}, v_{cc}) = \arg\max_{\substack{i=1,\ldots,N_\theta/2 \\ j=1,\ldots,N_\theta}} M_{cc}(i,j) \tag{27}$$

where the operator $\arg\max[\cdot]$ is used to obtain the coordinates of the maximum value in M_{cc}. Third, the H_C is adapted by the cyclic shift vector (u_{cc}, v_{cc}), which can be written as

$$H_C^{(i)\prime} = \psi\left(H_C^{(i)}, (u_{cc}, v_{cc})\right), \quad i \in [1, N_r] \tag{28}$$

where $H_C^{(i)\prime}$ is the ith matrix of the adapted H'_C. The feature arrays of all candidate points are adapted with the feature array of the reference point. Subsequently, these feature arrays of the reference point and the candidate points are transformed into vectorized descriptors. The Euclidean distances between reference and candidate descriptors are calculated. The pairwise reference and candidate points exhibiting the minimal distance between their corresponding descriptors are identified as the matches. Finally, FSC algorithm is performed to eliminate mismatches and obtain the final geometric transformation.

III. Experimental Results and Discussion

In order to validate the registration performance of the proposed method, six distinct categories of MRSI datasets were selected for qualitative and quantitative evaluation. Seven state-of-the-art (SOTA) registration methods were employed for comparative analysis, including ReDFeat [35], MINIMA-LG [36], RIFT2 [27], MS-HLMO [42], HOWP [43], POS-GIFT [29], and GLS-MIFT [30], all of which are specifically designed to address the multimodal characteristics. All implementations of the comparison methods were obtained from the publicly available code, and the optimal parameter settings were determined according to the recommendations provided by their authors.

A. Datasets

In this study, six real-world MRSI datasets were collected, including optical–NIR, NIR–SAR, optical–DSM, optical–infrared, and two optical–SAR datasets. The sample image pairs of the datasets are shown in Fig. 11. Each dataset comprises 20 image pairs captured across diverse scenes, yielding a total of 120 original image pairs. The detailed characteristics of each dataset are described as follows.

1) *Dataset 1:* This is an optical–NIR dataset derived from Sentinel-2 and Landsat 8. The dataset contains optical RGB and near-infrared (NIR) band remote sensing images. The images have 512 × 512 pixels in size.
2) *Dataset 2:* This is an NIR–SAR dataset derived from Landsat 9 and Sentinel-1. The dataset contains NIR band images and SAR images. The images have 512 × 512 pixels in size.
3) *Dataset 3:* This optical–DSM dataset is derived from the Potsdam and Vaihingen datasets. The Vaihingen dataset includes three-band IRRG (infrared, red, and green) and DSM image pairs, with images resized to 385 × 701–994 × 576. The Potsdam dataset contains four-band IRRGB (infrared, red, green, and blue) and DSM image pairs, with images resized to 512 × 512.
4) *Dataset 4:* This is an optical–infrared dataset derived from Sentinel-2 and Landsat 8. The dataset contains RGB remote sensing images and infrared band images. The images have 512 × 512 pixels in size.
5) *Dataset 5:* This is an optical–SAR dataset sourced from the OSdataset [44], comprising paired images with a size of 512 × 512 pixels.
6) *Dataset 6:* This is an optical–SAR dataset sourced from the WHU-OPT-SAR dataset. The original images in the WHU-OPT-SAR dataset were cropped to 512 × 512 pixels.

The image pairs include diverse terrain categories such as urban buildings and suburban landscapes. Furthermore, these image pairs exhibit prominent NIDs, which pose great challenges for image registration. Because the image pairs in these datasets are preregistered, their geometric transformation matrices are all identity matrices. To evaluate the rotation and scale invariance of the proposed method, the original

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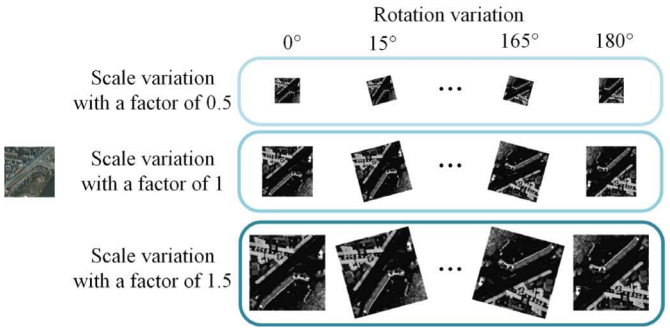


Fig. 12. Image pairs augmented with rotation and scale variations.

images were augmented with comprehensive transformations, including the rotation variations with the angles spanning 0°–180° at 15° intervals, along with the scale variations with the factors of 0.5, 1, and 1.5. The set of the image pairs with comprehensive transformations is illustrated in Fig. 12.

B. Evaluation Metrics and Parameter Settings

In our experiments, three quantitative metrics were employed for registration performance evaluation, i.e., NCMs, root-mean-square error (RMSE), and success rate (SR). NCM is defined as the number of matched point pairs where the location error is less than 3 pixels. The location error is calculated using the geometric transformation matrix **M**, which can be expressed as follows:

$$E = \sqrt{\|(\hat{x}_1, \hat{y}_1) - (x_2, y_2)\|^2} \tag{29}$$

$$\begin{bmatrix} \hat{x}_1 \\ \hat{y}_1 \\ 1 \end{bmatrix} = \mathrm{M} \cdot \begin{bmatrix} x_1 \\ y_1 \\ 1 \end{bmatrix} \tag{30}$$

where E denotes the location error, the terms (x_1, y_1) and (x_2, y_2) are the matched points on the reference and sensed images, respectively, and $(\hat{x}_1, \hat{y}_1)$ is the corresponding point of (x_1, y_1) on the sensed image transformed by the geometric transformation matrix **M**. RMSE is computed as the square root of the mean squared location errors for correct matches. SR is defined as the ratio of the number of image pairs with successful registration to the total number of image pairs, which can be calculated as follows:

$$\mathrm{SR} = \frac{1}{N_P} \sum_i I(P_i) \tag{31}$$

$$I(P_i) = \begin{cases} 1, & \mathrm{NCM}(P_i) \geq 10 \ \& \ \mathrm{RMSE} \leq 5 \\ 0, & \text{else} \end{cases} \tag{32}$$

where N_P denotes the total number of image pairs, P_i denotes the ith image pair, and $I(P_i)$ is a binary variable that equals 1 when the registration is successful and 0 when the registration fails. Here, the successful registration is determined by the combined judgment of NCM and RMSE. The term NCM(P_i) represents the NCM of the ith image pair. Higher values of NCM and SR, along with a lower value of RMSE, indicate better registration performance.

Parameters impose a significant impact on the robustness and accuracy of registration methods. With respect to the proposed method, three critical parameters prominently influence the registration performance, i.e., the radius of the local neighborhood R, the number of orientations N_θ, and the number of radial layers in the sampling grid N_r. Each parameter varied independently, whereas the others remained fixed, to which the corresponding results are presented in Table I. Here, NCM, RMSE and SR were calculated as the average results of the tested image pairs, which functioned as the evaluation metrics. We analyzed the impact of the parameters on the registration performance and determined their default values as follows.

TABLE I
AVERAGE RESULTS OF CSFD-AFM UNDER DIFFERENT PARAMETER SETTINGS

Test	Parameter setting			NCM	RMSE	SR(%)
	R	N_θ	N_r			
R	36	12	3	1514.37	2.08	93.27
	54	12	3	1604.28	1.95	96.15
	72	12	3	1627.08	**1.90**	**100.00**
	90	12	3	**1633.81**	1.92	**100.00**
	108	12	3	1618.62	1.94	98.40
N_θ	72	4	3	1543.39	2.09	95.19
	72	6	3	1594.93	1.96	**100.00**
	72	**12**	3	1627.08	1.90	**100.00**
	72	18	3	1635.72	**1.89**	**100.00**
	72	24	3	**1643.31**	**1.89**	**100.00**
N_r	72	12	1	1609.35	1.92	91.67
	72	12	2	1615.55	1.91	**100.00**
	72	12	**3**	**1627.08**	**1.90**	**100.00**
	72	12	4	1620.46	**1.90**	**100.00**
	72	12	5	1621.04	1.94	**100.00**

The parameter R denotes the radius of the local neighborhood. A radius R that is too small in the feature description causes insufficient information retention, resulting in poor distinctiveness. Conversely, an excessively large R expands the border regions (with a width of $R/2$) of the image shown in Fig. 3(c) and (f), thereby reducing the area of detectable feature points. When the image size is small, it may lead to the problem that feature points are concentrated in the center. As demonstrated in Table I, an increase in R causes a corresponding rise in NCM. Notably, when $R = 72$, the SR achieves 100% and the RMSE is minimized. In this study, we set $R = 72$ to realize a tradeoff between the descriptor distinctiveness and the uniform distribution of feature points. The parameter N_θ denotes the number of orientations in the steerable filter bank. Higher values of N_θ enhance the discriminability of orientations, thereby increasing the information capacity of the descriptors. It can be deduced from the experimental results in Table I that higher values of N_θ suggest more informative descriptors, which achieves higher NCM and lower RMSE

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TABLE II
AVERAGE RESULTS OF EIGHT METHODS ON ALL DATASETS

Metric	ReDFeat	MINIMA-LG	RIFT2	MS-HLMO	HOWP	POS-GIFT	GLS-MIFT	CSFD-AFM
NCM	6.13	42.65	14.36	114.18	5.30	435.74	1089.75	1658.46
RMSE	4.68	4.41	3.72	4.34	4.50	3.78	2.56	1.90
SR(%)	9.90	17.48	39.33	84.23	17.95	32.53	86.25	98.72

TABLE III
AVERAGE RESULTS OF EIGHT METHODS ON DIFFERENT DATASETS

Dataset	Metric	Method							
		ReDFeat	MINIMA-LG	RIFT2	MS-HLMO	HOWP	POS-GIFT	GLS-MIFT	CSFD-AFM
Optical-NIR	NCM	10.38	15.54	3.05	73.38	5.49	558.56	1521.69	**1673.31**
	RMSE	4.51	4.71	3.95	4.48	4.54	3.49	2.15	**1.91**
	SR(%)	14.36	13.35	28.21	69.23	20.51	38.46	92.83	**100**
NIR-SAR	NCM	2.18	34.36	5.77	76.69	1.28	252.15	1037.10	**1992.95**
	RMSE	4.74	3.74	4.25	4.57	4.76	4.57	2.53	**1.89**
	SR(%)	7.82	33.33	25.64	84.73	8.87	7.69	91.74	**100**
Optical-DSM	NCM	2.15	97.23	23.64	211.28	7.00	446.85	743.46	**1287.21**
	RMSE	4.75	4.57	3.28	4.27	4.37	3.16	2.54	**1.93**
	SR(%)	8.64	10.26	51.28	**100**	23.68	53.84	85.16	**100**
Optical-Infrared	NCM	1.64	3.36	4.90	97.72	5.90	206.28	849.77	**1519.31**
	RMSE	4.83	4.81	4.19	4.15	4.44	4.21	2.51	**1.97**
	SR(%)	5.13	5.84	22.59	76.92	17.95	6.67	87.18	**94.87**
Optical-SAR	NCM	3.05	26.59	11.23	123.79	3.41	380.33	975.62	**1647.36**
	RMSE	4.75	4.21	3.32	4.37	4.51	3.68	2.91	**1.87**
	SR(%)	10.52	23.08	51.84	92.31	21.32	46.15	78.54	**100**
Optical-SAR2	NCM	17.38	78.82	37.59	102.23	8.69	770.26	1410.85	**1830.59**
	RMSE	4.50	4.41	3.30	4.22	4.35	3.58	2.69	**1.83**
	SR(%)	12.95	19.03	56.41	82.17	15.38	42.35	82.05	**97.44**

results. However, this enlarges the size of the 3-D feature array, degrading its efficiency. Notably, when N increases beyond 12, both NCM and RMSE exhibit only marginal improvement.

Therefore, $N_\theta = 12$ is set as the default configuration. The parameter N_r represents the number of radial partitions of the local neighborhood. As illustrated in Table I, the maximum value of NCM and the minimum value of RMSE are simultaneously achieved when $N_r = 3$. As a result, $N_r = 3$ is determined as the default value. Based on the above results and analysis, we set the default parameters as $R = 72$, $N_\theta = 12$, and $N_r = 3$. We limited the maximum number of the detected feature points to 5000 and adopted them in the following experiments.

C. Evaluation of Rotation Invariance

In order to evaluate the rotation invariance of the proposed method, we conducted the experiments on the optical–NIR image pair, as shown in Fig. 11(a). The NIR image was rotated incrementally in 5° intervals over a full 360° rotation, yielding 72 discrete angles from 0° to 355°. The performance was quantitatively evaluated using NCM and RMSE, which were computed for each rotation angle. The RMSE was set to 5 pixels when the registration of an image pair failed. The comparison results with seven SOTA registration methods are presented in Fig. 13.

As shown in Fig. 13(a), NCM under varying rotation angles is presented. ReDFeat and MINIMA-LG failed to establish correct matches beyond 30° of rotation, due to the lack of dedicated modules for handling large-angle rotations. The HOWP showed markedly degraded performance under rotation, indicating limited robustness. The RIFT2 produced a limited NCMs and exhibited numerous matching failures. Although the POS-GIFT obtained a considerable NCMs, its performance was unstable with a high failure rate. The MS-HLMO consistently produced correct matches across all rotation angles with

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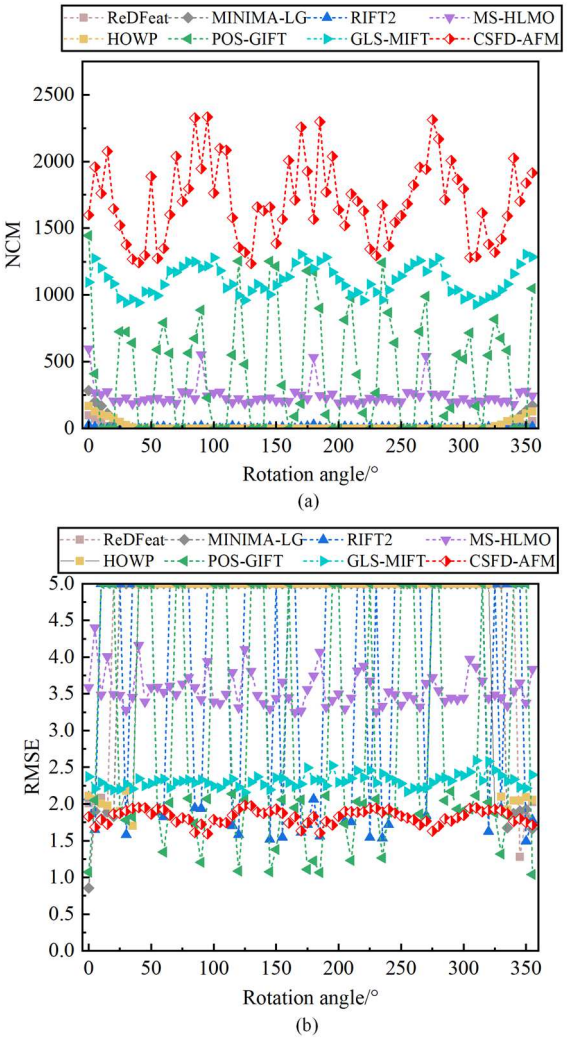


Fig. 13. Registration performance of different methods under different rotation angles. (a) NCM results. (b) RMSE results.

limited numbers. The GLS-MIFT achieved the second-highest NCM, second only to the proposed method. In contrast, the proposed approach consistently outperformed all other methods in NCM across all rotation angles. In terms of accuracy, the RMSE results are presented in Fig. 13(b). The ReDFeat, MINIMA-LG, and HOWP failed when confronted with rotation angles exceeding 30°. While the RIFT2 and POS-GIFT occasionally yielded slightly higher accuracy, they were prone to frequent failures at numerous other angles. The MS-HLMO, GLS-MIFT, and CSFD-AFM achieved successful registration across various angles. However, the RMSE values obtained by CSFD-AFM were lower than those of MS-HLMO and GLS-MIFT. The RMSE values obtained by CSFD-AFM remained consistently below 2 pixels, suggesting that the CSFD-AFM effectively balanced robustness and accuracy for image registration under arbitrary rotations.

D. Qualitative and Quantitative Results

The output matches of the ReDFeat, MINIMA-LG, RIFT2, MS-HLMO, HOWP, POS-GIFT, GLS-MIFT, and our proposed CSFD-AFM are presented in Figs. 14–17. To evaluate performance under different geometric distortions, image pairs were categorized into four groups: translation, rotation, scaling, and composite transformation (combining scaling, translation, and rotation). Due to their significant NIDs and geometric distortions, these image pairs pose great challenges for registration methods.

As shown in Fig. 14, the output matches of SOTA methods and CSFD-AFM on translated images were presented. The ReDFeat and MINIMA-LG degraded substantially when applied to optical–SAR image pairs with significant NIDs, suggesting their limited modality-invariant capabilities. Although the POS-GIFT performed effectively on most image pairs, its performance deteriorated significantly on the optical–SAR image pair, where it generated numerous incorrect matches due to its limited stability. The performance of RIFT2 and HOWP degraded significantly when applied to optical–infrared and NIR–SAR image registration tasks, respectively. The MS-HLMO, GLS-MIFT, and CSFD-AFM consistently produced correct matches across all six scenes, demonstrating their modality-invariant properties.

As shown in Fig. 15, the output matches of SOTA methods and CSFD-AFM on scaled images are presented. The deep learning methods ReDFeat and MINIMA-LG exhibited significant performance variations across different scenarios. The RIFT2 was sensitive to scale changes due to its lack of scale space construction, and failed for all image pairs except the optical–DSM one. While the HOWP failed for NIR–SAR and optical–SAR2 image pairs, it achieved only a limited NCMs in other scenarios. The MS-HLMO, POS-GIFT, and GLS-MIFT yielded correct matches across all scenarios. However, the NCMs decreased for the MS-HLMO and POS-GIFT in optical–DSM and optical–infrared image pairs, respectively, and for the GLS-MIFT when the scale factor exceeded 1. In contrast, the proposed CSFD-AFM consistently generated a large number of uniformly distributed correct matches across all scenes with scale variations, demonstrating its pronounced scale invariance.

As shown in Fig. 16, the output matches of SOTA methods and the CSFD-AFM on rotated images are presented. The deep learning methods ReDFeat and MINIMA-LG failed in rotated images, as they lack explicit mechanisms to handle large rotational transformations. While the RIFT2 and HOWP yielded a limited NCMs in optical–DSM and optical–NIR image pairs, respectively, they failed completely in other modalities. Although the POS-GIFT produced a considerable NCMs for optical–NIR, optical–DSM, and optical–SAR2 image pairs, it yielded a high number of incorrect matches in other image pairs. The proposed CSFD-AFM demonstrated superior overall performance compared with the MS-HLMO and GLS-MIFT, achieving a higher NCMs with better spatial distribution.

As shown in Fig. 17, the output matches of SOTA methods and the CSFD-AFM on images with composite transformations are presented. When dealing with challenging images involving composite transformations, the ReDFeat, MINIMA-LG, RIFT2, and HOWP failed to generate correct matches. NCMs for the MS-HLMO, POS-GIFT, and GLS-MIFT was

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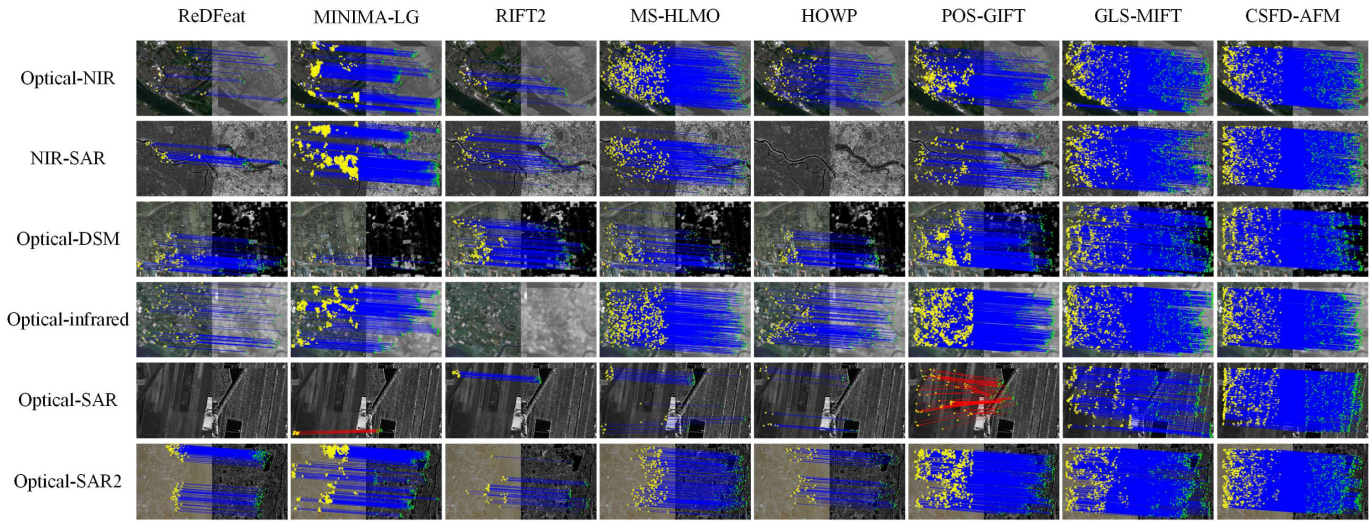


Fig. 14. Matching results produced by each method for image pairs under translational variations. The blue and red lines represent the correct and incorrect matches, respectively.

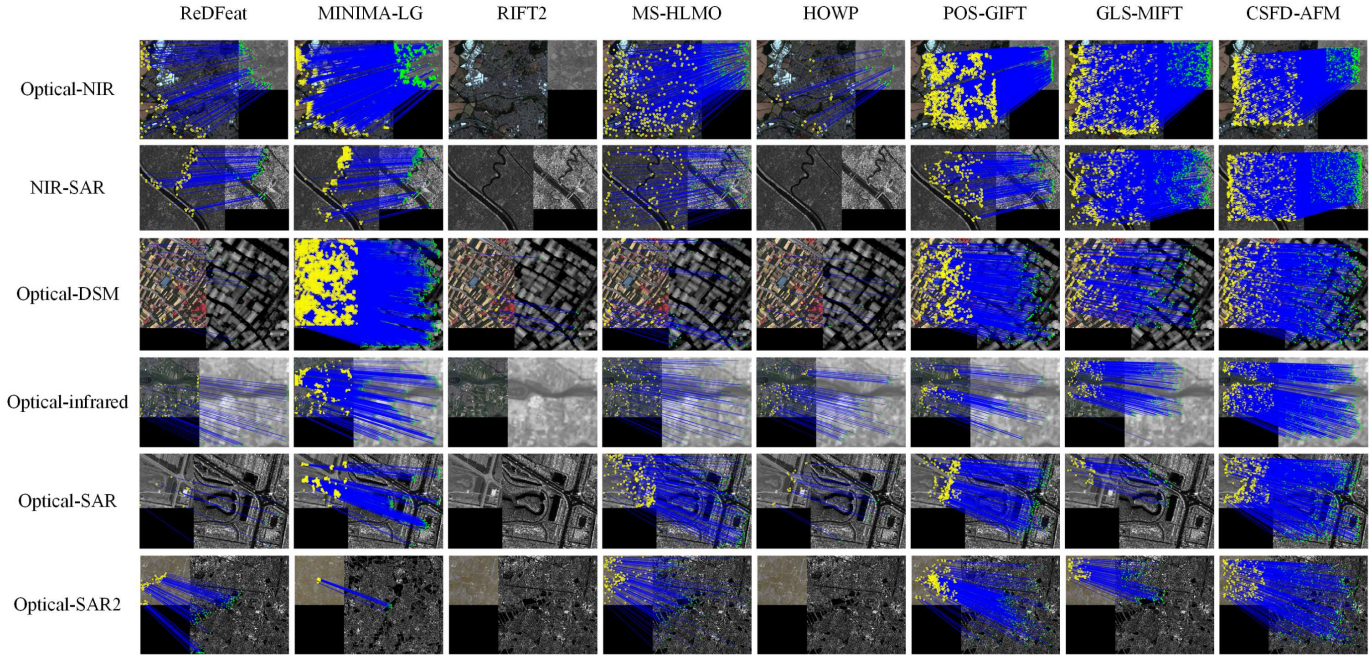


Fig. 15. Matching results produced by each method for image pairs under scaling variations. The blue and red lines represent the correct and incorrect matches, respectively.

relatively small, with a nonuniform distribution, and there were many invalid cases. In contrast, our proposed CSFD-AFM demonstrated superior performance by consistently producing a dense and spatially uniform set of accurate correspondences across all tested scenarios. This result verifies its robust capability in handling complicated composite transformations.

Table II presents the average results of the CSFD-AFM and SOTA methods on all datasets for various metrics. Table III presents the results (NCM, RMSE, and SR) for each dataset. Each value represents the average result for image pairs in the corresponding dataset. Fig. 18 illustrates the performance comparison between the proposed CSFD-AFM and SOTA methods.

Although the deep learning-based methods ReDFeat and MINIMA-LG are capable of matching multimodal features, their performances on our datasets were strictly limited due to the lack of robustness to large rotation variations. They were not adequately designed or trained to deal with rotation variations between images. Consequently, the average SR values were 9.90% and 17.48%, the average NCM scores were 6.13 and 42.65, and the average RMSE reached 4.68 and 4.41 pixels, respectively.

The RIFT2 addresses NIDs in multimodal images by constructing descriptors on the MIM. However, the RIFT2 achieved an average SR of 39.33%, with an average RMSE of 3.72. The RIFT2 yielded significantly low NCM values, with

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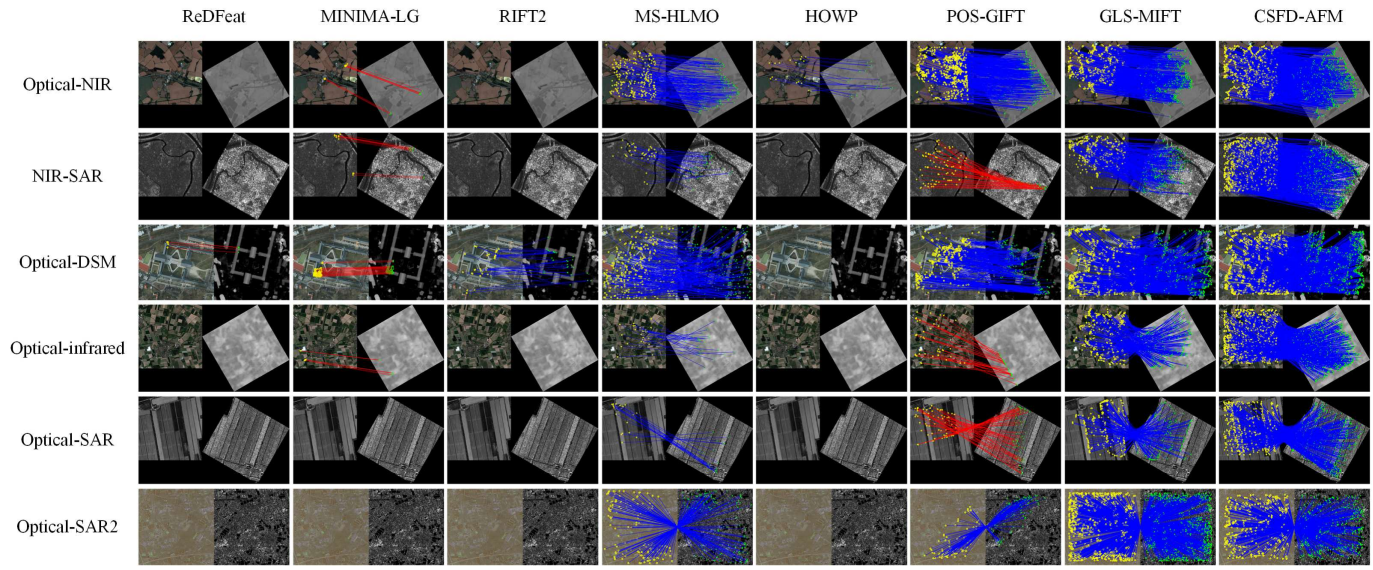


Fig. 16. Matching results produced by each method for image pairs under rotation variations. The blue and red lines represent the correct and incorrect matches, respectively.

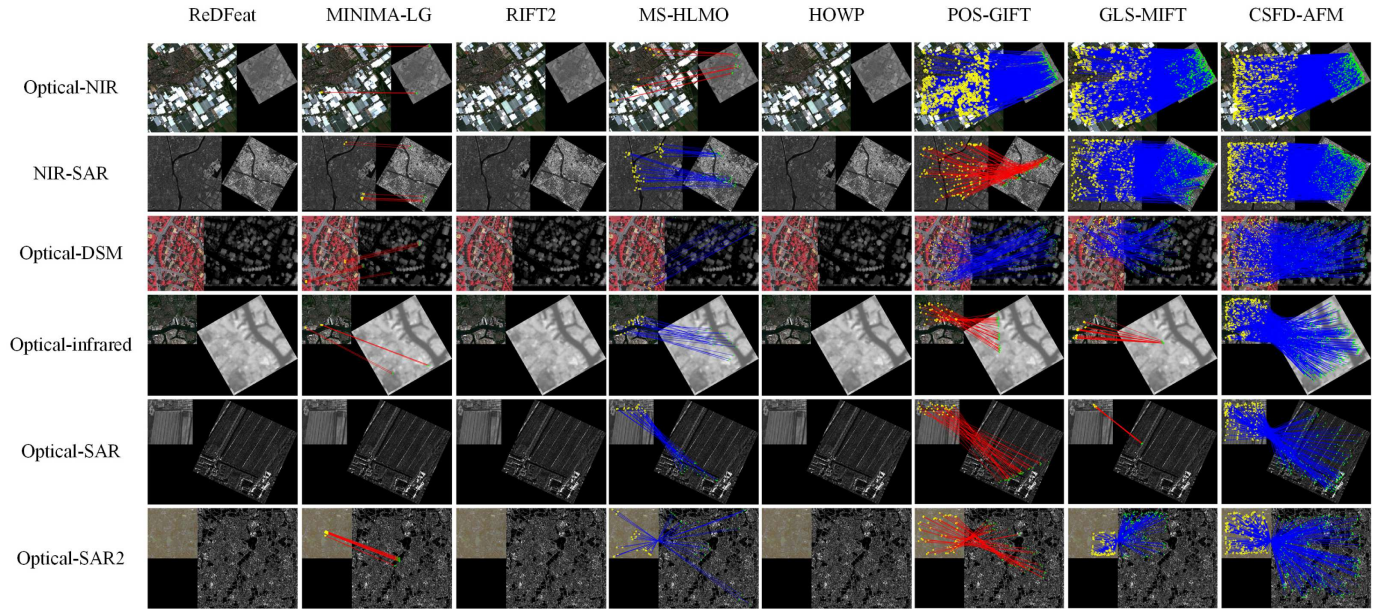


Fig. 17. Matching results produced by each method for image pairs under composite variations. The blue and red lines represent the correct and incorrect matches, respectively.

a maximum value of only 37.59 across all datasets. This is primarily because the RIFT2 lacks a dedicated mechanism for handling scale variations.

The MS-HLMO achieved a relatively high average SR of 84.23%. However, it achieved an average NCM of 114.18 and an average RMSE of 4.34. The relatively low performance in NCM and RMSE stems from inadequate feature point detection in the original images.

The HOWP achieved an average SR of 17.95%, with an average NCM of 5.30 and an average RMSE of 4.50. The main drawback of the HOWP is its instability under large-angle rotations, which significantly compromises its overall performance.

The POS-GIFT achieved an average NCM of 435.74, with an average RMSE of 3.78. However, the POS-GIFT demonstrated a relatively low SR, varying considerably from as low as 6.67% to merely 53.84%. This significant discrepancy is primarily attributed to the registration failures at specific orientation angles, demonstrating its limited rotation invariance.

The GLS-MIFT achieved the performance comparable with that of the proposed CSFD-AFM, with an average NCM of 1089.75, an average RMSE of 2.56 pixels, and an average SR of 86.25%. Despite consistently achieving an RMSE below 3 pixels, the RMSE of GLS-MIFT varied considerably across the datasets. A pronounced SR drop was evident on the

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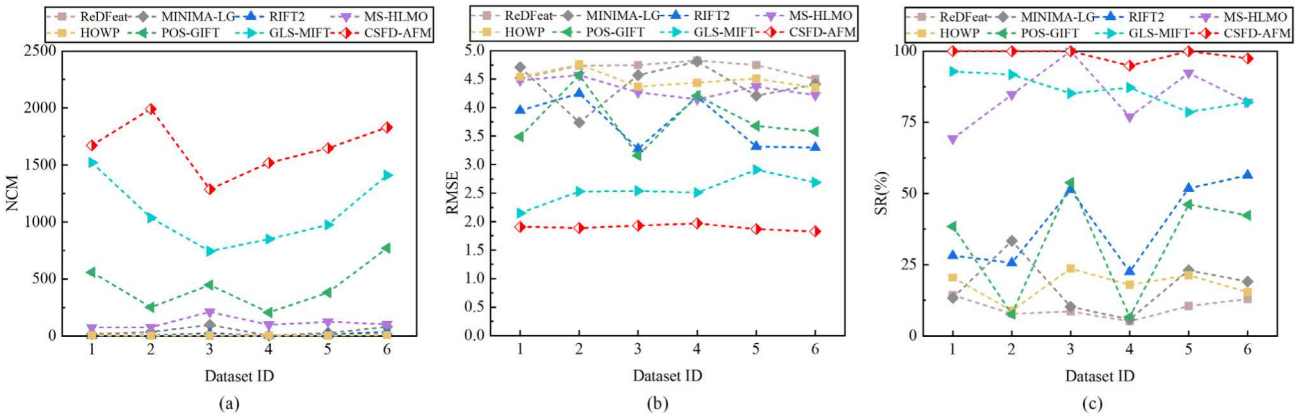


Fig. 18. Performance comparison of different registration methods. (a) NCM results. (b) RMSE results. (c) SR results.

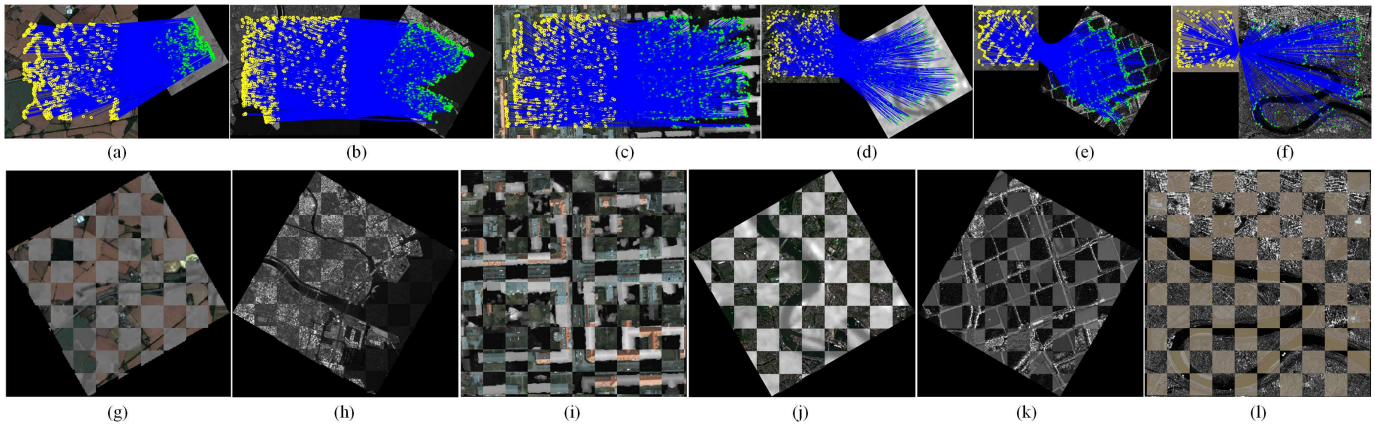


Fig. 19. Correct matches and checkerboard visualizations of CSFD-AFM. (a)–(f) Correct matches for Optical-NIR, NIR-SAR, Optical-DSM, Optical-Infrared, Optical-SAR, and Optical-SAR2 pairs, respectively. (g)–(l) Corresponding checkerboard visualizations for (a)–(f).

TABLE IV
AVERAGE RUNNING TIME RESULTS ON SIX DATASETS

Method	ReDFeat	MINIMA-LG	RIFT2	MS-HLMO	HOWP	POS-GIFT	GLS-MIFT	CSFD-AFM
T(s)	1.02	4.46	12.34	62.23	5.42	18.77	20.36	16.21

more challenging optical–SAR and optical–SAR2 datasets, demonstrating its limitations in handling SAR images.

Across all six datasets, the proposed CSFD-AFM method achieved the highest average NCM of 1658.46, with an average RMSE of 1.90 pixels. The CSFM-AFM method achieved a 100% SR across optical–NIR, NIR–SAR, optical–DSM, and optical–SAR datasets. However, due to the absence of distinct structural features in infrared images, the CSFD-AFM failed to register all images of the optical–infrared dataset successfully, yielding an SR of 94.87% and a marginally higher RMSE of 1.97. Nevertheless, it still outperformed all SOTA methods. Although there are challenges such as cloud-induced blurring in the optical–SAR2 dataset, which occasionally led to registration failures, our method attained an SR of 97.44%, significantly exceeding the 82.05% SR achieved by the GLS-MIFT.

Fig. 19 presents correct matches and checkerboard visualizations of the CSFD-AFM. The image pairs from each multimodal dataset were subjected to different scale and rotation variations. Fig. 19(g)–(l) presents the corresponding registration results of Fig. 19(a)–(f), displayed using checkerboard maps. These maps were generated from the transformation matrices derived from the correct matches. Most image pairs are accurately aligned, with ground features exhibiting continuous outlines. The edges in the checkerboard maps align well without obvious misalignment, further demonstrating the considerable accuracy of the proposed CSFD-AFM.

E. Average Running Time

Table IV presents the average running time (T) results for the ReDFeat, MINIMA-LG, RIFT2, MS-HLMO, HOWP, POS-GIFT, GLS-MIFT, and CSFD-AFM. All experiments were conducted under the same configuration, using a 12th Generation Core i5-12600KF CPU and an RTX 4060Ti GPU.

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Although both of the learning-based methods, ReDFeat and MINIMA-LG, leverage CUDA for acceleration and achieve lower computation times, their suboptimal performance in rotational scenarios significantly undermines the practical value of this efficiency. The HOWP achieved the fastest runtime among all handcrafted methods. However, its SR remained comparatively low. MS-HLMO exhibited the longest computation time at 62.239 s among all algorithms due to its computationally intensive Gaussian pyramid construction. The RIFT2 avoids scale-space construction, leading to higher computational efficiency compared to the POS-GIFT, GLS-MIFT, and CSFD-AFM. POS-GIFT, GLS-MIFT, and CSFD-AFM all adopted frameworks with a two-step matching process. While POS-GIFT and GLS-MIFT demonstrated comparable time costs, the proposed CSFD-AFM achieved a shorter computation time due to its efficient assignment of multioriented descriptors to feature points via cyclic shifting. Although the two-step matching methods yield better matching results, their computational efficiency suggests considerable room for optimization. Therefore, the use of parallel processing techniques shall be considered to improve efficiency.

F. Limitations and Future Work

Our proposed CSFD-AFM method constructs feature descriptors by extracting multiscale and multiorientation structural features from images using steerable filters. As shown in Table III, the optical–infrared dataset presents greater challenges for the CSFD-AFM, primarily due to the relatively low resolution of the infrared images, which limits the availability of distinct features. Similarly, the presence of thin haze in optical images, which causes blurring and reduced contrast, adversely affects the registration performance on the optical–SAR2 dataset. In future work, we will focus on enhancing image preprocessing techniques to improve the extraction of structural features.

IV. Conclusion

In this study, we propose an ingenious MRSI registration method based on CSFD and AFM to address severe NIDs and geometric distortions. In the registration framework, the feature points are first detected in scale spaces. Subsequently, a 3-D feature array is constructed on the HMIM, followed by a cyclic shift operation based on the dominant orientation. The array is then subjected to dimensionality reduction and vectorization to form the CSFD. Subsequently, the AFM strategy is implemented. Specifically, the initial transformation derived from CSFDs is utilized to adapt the configuration, regenerate feature arrays, and constrain the matching scope. With the narrowed matching range, a cross-correlation-based cyclic shift method is employed for orientation alignment, resulting in the adjusted CSFDs. We evaluated the registration performance of the proposed CSFD-AFM on MRSIs augmented with various geometric variations. The experimental results demonstrate that compared with seven SOTA methods, our proposed CSFD-AFM achieves advantageous robustness and precision under severe geometric distortions across different modalities. Although the proposed method provides robust registration performance, it is still challenging for the method to deal with the images deficient in distinct structural features. In future work, we will be focusing on extracting structural features from multimodal images degraded by low resolution and haze.

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